



United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4891: Data Mining

Midterm Exam : Fall 2025 Time: 1 Hour 30 Minutes Marks: 30

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Answer all of the following questions sequentially.

1. Given the dataset in Table 1. Calculate the probability of playing or not playing for the temperature of 33°F. Also, calculate the temperature with a probability 91% for play. The parameter v for the activation function is given as $v = -27 + 3 * \text{temperature}$. [You are allowed to apply an appropriate regression method] [6]

Table 1

Temperature °F	Play
65	No
56	No
37	Yes
42	Yes
24	Yes

2. a) Estimate the Simple Matching and Jaccard coefficients from the following binary patterns. [4]

p = 1 0 0 1 0 0 0 0 0 1
q = 0 0 0 1 0 0 1 0 0 1 0

- b) Measure the distance matrix for the following data points through an appropriate similarity and dissimilarity measures with parameter $r=2$. [4]

Data points	x	y
P1	1	2
P2	2	1
P3	4	2
P4	6	3

3. a) Discuss the dice OLAP operation with a pictorial example. [2]

- b) You are working on a 3-class classification problem (classes: A, B, and C). After validation of your model, you observed the following confusion matrix: [6]

Classes	Actual classes			
	----	A	B	C
Predicted classes	A	50	10	5
	B	9	61	12
	C	5	15	55

Calculate the specificity and draw the ROC curve along with each class.

4. You get the following data set. You are trying to determine whether a customer finds a particular type of food appealing based on the food's temperature, taste, and size. [2+6]

- i) What is the initial entropy of the appealing class?
ii) Draw a decision tree using an appropriate algorithm.

Appealing	Temperature	Taste	Size
No	Hot	Salty	Small
No	Cold	Sweet	Large
No	Cold	Sweet	Large
Yes	Hot	Sour	Small
No	Hot	Salty	Large
Yes	Hot	Sour	Large
Yes	Cold	Sweet	Small
Yes	Cold	Sweet	Small